

# Environmental assessment of electricity production in Tanzania

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## ARTICLE INFO

### Article history:

Received 25 August 2011

Revised 25 July 2012

Accepted 25 July 2012

Available online 19 August 2012

### Keywords:

Electricity

Emissions

Energy

Environment

Generation

Life cycle assessment

## ABSTRACT

The use of fossil fuels for electricity generation contributes to local air pollution as well as to the greenhouse effect. Electricity generation technologies contribute to environmental impacts including acidification, eutrophication, resource depletion, and climate change. This paper presents the environmental assessment of the centralized grid-connected electricity production in Tanzania using a life cycle approach for the years 2000, 2015, 2020, 2026 and 2030, according to the Tanzania Electricity Supply Company Limited, TANESCO's, plans for power system expansion (power system master plan of the year 2009). The results of this study show that in all the chosen impacts categories, based on the absolute annual electricity production, environmental impacts increase significantly with time when comparing the years 2000, 2015, 2020, 2026 and 2030.

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## Introduction

This paper presents the results of the environmental impacts that are likely to occur as a result of electricity generation from hydropower and fossil-based fuel sources viz. coal, natural gas, industrial diesel oil (IDO) and heavy fuel oil (HFO). An earlier study highlighted the environmental effects that would probably occur as a result of the over exploitation of forest resources for biomass use in Tanzania (Felix and Gheewala, 2011).

Electricity supply is a major socio-economic problem in Tanzania. With a population of about 43 million people and with about 80% of the population living in rural areas, only 14% of the total population in 2009 had access to electricity (GoT, 2010; The World Bank, 2012). In 2001, about 10% of the population had access to electricity (Odhiambo, 2009), of which about 2% of rural inhabitants were connected to the national grid. While the country has the lowest electrification levels in the world, its electricity demand is growing at the rate of 9–10% each year (Mbendi, 2005; TANESCO, 2010).

The installed capacity of power plants increased from 782 MW in 2000 to 1051 MW in 2009 (Mwakapugi et al., 2010). However, most of the existing generation capacity is not in full operation, so that generation is often below the peak demand of 900 MW. The low development in this sector is mainly due to the lack of capital for developing new power plants and poor maintenance of the existing power plants. In addition to the above factors, the national grid has high transmission and distribution losses, though losses have fallen in recent years (Table 1) (TANESCO, 2010; The World Bank, 2012).

**Table 1**

Transmission and distribution losses (% output) of the national grid (The World Bank, 2012).

| Year   | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|--------|------|------|------|------|------|------|------|------|------|------|
| Losses | 22%  | 25%  | 24%  | 27%  | 20%  | 27%  | 26%  | 19%  | 19%  | 19%  |

Tanzania's energy sector is characterized by low per-capita consumption of commercial energy such as kerosene, petroleum and a large dependence on biomass energy in form of firewood, charcoal and agricultural waste as well as human and animal waste. In 1990, the total biomass resource potential was 27 million tonnes of oil equivalent (Mtoe) from the natural forests (Sheya and Mushi, 2000).

The energy balance of Tanzania shows that biomass use accounts for over 90% of energy consumption and continues to dominate as the major source of energy. Petroleum and electricity account for about 8% and 1.2% respectively (Odhiambo, 2009). The use of relatively expensive generation technology in isolated grids leads to relatively high costs for electricity supply which is why some of the households prefer connectivity to a centralized grid. Since many parts of the country are not connected to the national grid, therefore kerosene is the most widely used fuel for lighting, and consequently increasing kerosene prices pose an additional burden on many households.

The Tanzania Electric Supply Company (TANESCO) is the organization responsible for electricity generation, transmission and distribution. The company also conducts research for future development plans in electricity generation to meet the country's requirements. The country's power generation system includes the use of hydro, thermal

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**Table 2**

Total electricity generation during 2000–2007 (CD4CDM, 2010; ES, 2009; TANESCO, 2010; TSED, 2010).

| Year              | 2000 | 2001 | 2002 | 2003 | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 |
|-------------------|------|------|------|------|------|------|------|------|------|------|
| Production (GW h) | 2412 | 2668 | 2720 | 2587 | 2147 | 3514 | 3422 | 4728 | 2917 | 3290 |

and gas power with hydropower contributing about 90% of the total electricity generation in 2008. The government has enacted the new electricity act in 2008 which encourages participation of the private sectors in generation and distribution of electricity which thus reduces TANESCO's monopoly and increases competition (TANESCO, 2010).

Total electricity generation during 2000–2009 is shown in Table 2; with a large share from the hydropower stations (TSED, 2010). Mtera (80 MW), Kihansi (180 MW) and Kidatu (204 MW) dams are the largest hydropower plants. Mtera dam is the most important reservoir in the power generation system because it can provide a storage capability of over a year. These hydroelectric plants are built on rain fed rivers (Kainkwa, 1999; Mbendi, 2005). Much of the resource potential is still untapped. The hydropower potential is estimated to be about 4700 MW, of which 10–12% has so far been developed. Also, the country has limited fossil fuel resources with proven reserves of 304 million tonnes of coal and 45 billion m<sup>3</sup> of natural gas (Byakola et al., 2009; Kabaka and Gwang'ombe, 2007; Olotu, 2010).

The electricity generation through the national grid and other sources varies not only due to the uneven distribution of rainfall throughout the year to run hydro turbines in hydropower stations, poor maintenance and bad conditions of the existing thermal power

plants units, but also the lack of “proper institutional rules and regulations, and financial mechanisms to make markets work in support of energy for the sustainable development” (UNDP, 2004). For example in 2007, total electricity production was 4728 GW h, decreasing to 2917 GW h in 2008, and then increasing to 3290 GW h in 2009 (Table 2). The increase in electricity production in 2008 to 2009 equivalent to 11% is due to the increase in electricity production from natural gas fired power plants in the grid mix (ES, 2009).

### Goal and scope of the study

The aim of this study is to assess the environmental burdens associated with different electric power production systems in Tanzania over the period from 2000 to 2030. In addition, the study aims to evaluate the status of the available resources for the electricity production, and the projection of the electricity production in the future. Types of power plants chosen are natural gas fired power plants, coal fired power plants, hydropower plants, heavy fuel oil fired power plants and industrial diesel oil fired power plants.

This study presents a cradle to gate life cycle assessment (LCA) comprising the extraction and processing of fuels, transport of the fuels to the power plants and electricity generation from the power plants as shown in Fig. 1. This study assumes that coal and natural gas are mined and extracted in Tanzania while heavy fuel oil and industrial diesel oil are imported.

This study takes into account the building of infrastructure (including headponds, transmission and distribution equipment) of hydropower plants for it has significance in the emission of greenhouse gases (Frischknecht et al., 2007; Ruether et al., 2004; von Uthmann et al., 2005). The building of infrastructure for natural gas, coal and oil fired power plants is also included but its contribution to the life cycle emis-

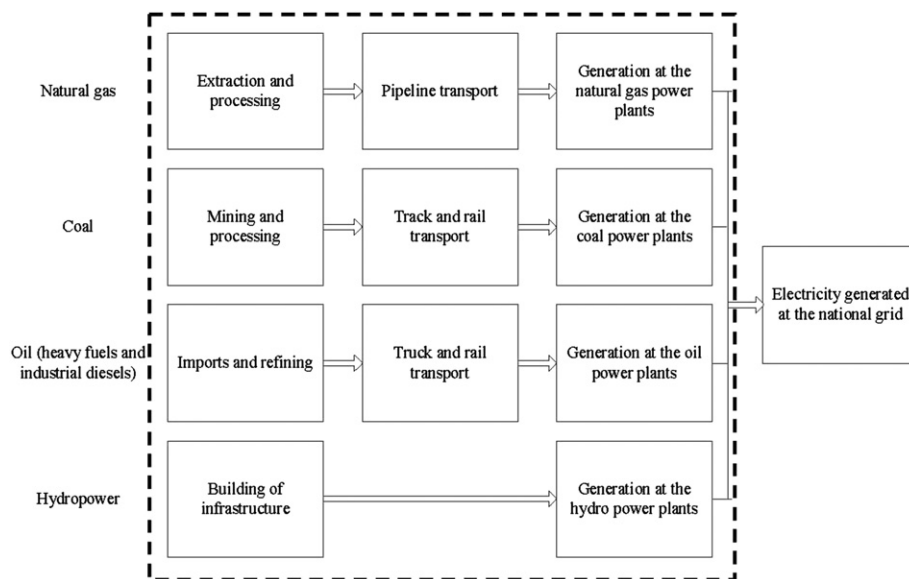


Fig. 1. Scope of the life cycle assessment of electricity production.

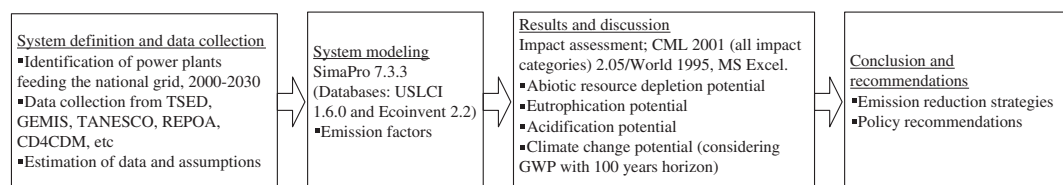


Fig. 2. Methodology overview for assessing environmental impacts of electricity production in Tanzania.

**Table 3a**

Power system master plan (CD4CDM, 2010; Mwakapugi et al., 2010; TANESCO, 2010).

| Project                                       | Resource/<br>technology | Planned<br>capacity (MW) | Year completed/to<br>be completed |
|-----------------------------------------------|-------------------------|--------------------------|-----------------------------------|
| Tegeta                                        | Gas/GE                  | 45                       | 2009                              |
| Conversion of IPTL from<br>HFO to natural gas | Gas/GE                  | 100                      | 2011                              |
| Kiwira, phase I                               | Coal/FBS                | 200                      | 2011                              |
| Nyakato/Mwanza                                | HFO/FOE                 | 60                       | 2011                              |
| Ubungo                                        | Gas/GT                  | 100                      | 2011                              |
| Kiwira, phase II                              | Coal/FBS                | 200                      | 2012                              |
| Singida                                       | Wind/WT                 | 50                       | 2012                              |
| Zambia–Tanzania<br>interconnector             | –                       | 200                      | 2012                              |
| Kinyerezi                                     | Gas/CC                  | 240                      | 2013                              |
| Rusumo Falls (62 MW,<br>33% for Tanzania)     | Hydro/ROR               | 21                       | 2015                              |
| Somanga Funga (210 MW in<br>2014 as OC)       | Gas/CC                  | 310                      | 2016                              |
| Malagarasi                                    | Hydro/ROR               | 8                        | 2016                              |
| Ruhudji                                       | Hydro/RES               | 358                      | 2016                              |
| Mnazi Bay                                     | Gas/CC                  | 300                      | 2017                              |
| Mtwara Artumas                                | Gas/GE                  | 12                       | 2017                              |
| Rumakali                                      | Hydro/RES               | 222                      | 2018                              |
| Stigler's Gorge, phase I                      | Hydro/RES               | 300                      | 2020                              |
| Stigler's Gorge, phase II                     | Hydro/RES               | 600                      | 2023                              |
| Mchuchuma, phase I                            | Coal/FBS                | 200                      | 2023                              |
| Mchuchuma, phase II                           | Coal/FBS                | 200                      | 2024                              |
| Ngaka                                         | Coal/FBS                | 400                      | 2024                              |
| Stigler's Gorge, phase III                    | Hydro/RES               | 300                      | 2026                              |
| Kakono                                        | Hydro/RES               | 53                       | 2027                              |
| Nyasa                                         | Coal/FBS                | 200                      | 2027                              |
| Masigira                                      | Hydro/RES               | 118                      | 2028                              |
| Mpanga                                        | Hydro/RES               | 144                      | 2028                              |
| Ikondo–Mnyera                                 | Hydro/RES               | 340                      | 2029                              |
| Taveta–Mnyera                                 | Hydro/RES               | 145                      | 2030                              |

IPTL—Independent Power Tanzania Limited, HFO—Heavy Fuel Oil, FBS—Fluidized Bed Combustor, CC—Combined Cycle, ROR—Run of River, RES—Reservoir, FOE—Fuel Oil Engine, GT—Gas Turbine, OC—Open Cycle, WT—Wind Turbine, and GE—Gas Engine.

**Table 3b**

Other power stations studied (CD4CDM, 2010; Mwakapugi et al., 2010; TANESCO, 2010).

| Power<br>stations   | Resource/<br>technology | Installed capacity<br>(MW) | Year of<br>commission |
|---------------------|-------------------------|----------------------------|-----------------------|
| Hale                | Hydro/ROR               | 21                         | 1967                  |
| Nyumba ya<br>Mungu  | Hydro/RES               | 8                          | 1968                  |
| Kidatu              | Hydro/RES               | 204                        | 1975                  |
| Ubungo              | IDO/DE                  | 34.4                       | 1980                  |
| Nyakato/<br>Mwanza  | IDO/DE                  | 12.5                       | 1980                  |
| Zuzu/Dodoma         | IDO/DE                  | 7.4                        | 1980                  |
| Iyunga/Mbeya        | IDO/DE                  | 13.9                       | 1980                  |
| Musoma              | IDO/DE                  | 2.6                        | 1980                  |
| Tabora              | IDO/DE                  | 11                         | 1980                  |
| Mtera               | Hydro/RES               | 80                         | 1988                  |
| Pangani falls       | Hydro/ROR               | 68                         | 1995                  |
| Kihansi             | Hydro/RES               | 180                        | 2000                  |
| Ubungo EPP          | Gas/GT                  | 100                        | 2000                  |
| IPTL/Tegeta         | HFO/FOE                 | 100                        | 2002                  |
| Songas              | Gas/GT                  | 120                        | 2004                  |
| Ubungo/<br>AGGREKO  | Gas/GT                  | 40                         | 2000                  |
| Dowans              | Gas/GT                  | 35                         | 2007                  |
| Ubungo gas<br>plant | Gas/GE                  | 102.5                      | 2007                  |
| ALSTHOM/<br>Mwanza  | HFO/FOE                 | 40                         | 2007                  |
| Nyakato/<br>Mwanza  | HFO/FOE                 | 60                         | 2011                  |

IDO—Industrial Diesel Oil, DE—Diesel Engine, EPP—Emergency Power Project.

**Table 4**

Installed capacity of the power plants according to the power system master plan, 2000–2030 (CD4CDM, 2010; Mwakapugi et al., 2010; TANESCO, 2010; TSED, 2010).

| Installed capacity of the power plants (MW) | 2000 | 2015 | 2020 | 2026 | 2030 |
|---------------------------------------------|------|------|------|------|------|
| Hydro                                       | 561  | 582  | 1470 | 2370 | 3170 |
| Natural gas                                 | 140  | 1038 | 1450 | 1450 | 1450 |
| Industrial diesel oil (IDO)                 | 81   | 81   | 81   | 81   | 81   |
| Heavy fuel oil (HFO)                        | 0    | 60   | 60   | 60   | 60   |
| Coal                                        | 0    | 400  | 400  | 1200 | 1400 |
| Total                                       | 782  | 2160 | 3460 | 5160 | 6160 |

sions is usually less relevant; operation is much more relevant, that is to say only emissions due to mining (extraction) and combustion of coal, natural gas and oil at the plant are significant (Frischknecht et al., 2007; Ruether et al., 2004). Electricity stakeholders may use the results of this study to have an idea about the possible environmental consequences of current electricity production methods (production systems and technologies). Some suggestions on improving the environmental profile of electricity production are also presented.

The country has limited traditional fossil fuel resources and the economy depends on rain-fed agriculture (Byakola et al., 2009). Therefore, potential increase in global temperature (leading to complex effects on climate), eutrophication and acidification would have adverse impacts on environment, energy resources (energy security), agriculture (food security), and therefore on the country's economy (Guinée et al., 2002). The eutrophying and acidifying substances emitted by power plants can be transported by rainfall; therefore, they would have impact on agriculture. In the above context, the environmental impacts of relevance are abiotic resource depletion potential (ADP), climate change potential (CCP), acidification potential (AP), and eutrophication potential (EP).

The authors of this paper strongly suggest that it is important to study electricity because it helps many things to function such as (high efficiency) electric lamps, electronic equipment (including radio, television, and mobile phones), electric fans, refrigerators, air-conditioners, etc. The use of electricity in productive activities e.g. processing of agricultural products, industries, etc helps to boost (industrial development, economic development and create wealth (Ferguson et al., 2000) thus alleviating poverty and empowering the poor i.e. job creation, increase productivity and standard of living (Burney, 1995).

Sustainable development means that “the present generations meet their current needs without compromising the ability of the future generations to meet their needs” (WCED, 1987). The development process should include a reliable and continuous electricity supply to meet the country's demand at affordable costs to the majority and with the growing concerns over the anthropogenic climate change, and the implications of local pollution on human health (Sakulniyomporn et al., 2011) and the impact of both climate change and pollution on ecosystems. Therefore, the overall environmental performance of electricity generation systems should be studied in the context of sustainable development.

Since in most places, electricity is generated by burning of limited fossil fuels resources (coal, natural gas, heavy fuel oils), therefore, it is necessary to use electricity wisely, because there is a probability of having not enough resources for the electricity generation in the

**Table 5**

Electricity production (GW h) according to the power system master plan (2000–2030).

| Year                  | 2000 | 2015   | 2020   | 2026   | 2030   |
|-----------------------|------|--------|--------|--------|--------|
| Coal                  | 0    | 2600   | 2600   | 7800   | 9100   |
| Natural gas           | 216  | 6744   | 9422   | 9422   | 9422   |
| Heavy fuel oil        | 0    | 390    | 390    | 390    | 390    |
| Industrial diesel oil | 51   | 526    | 526    | 526    | 526    |
| Hydro                 | 2145 | 2619   | 6615   | 10,665 | 14,265 |
| Total                 | 2412 | 12,879 | 19,553 | 28,803 | 33,703 |

**Table 6**  
Efficiency factors for different types of power plants (UNFCCC, 2011 and TANESCO).

| Power plant type | Technology                    | Efficiency (%) |
|------------------|-------------------------------|----------------|
| Natural gas      | Gas engine (GE)               | 45             |
| Natural gas      | Gas turbine (GT)              | 42             |
| Natural gas      | Combined cycle (CC)           | 60             |
| Natural gas      | Open cycle (OC)               | 39.5           |
| Coal             | Fluidized bed combustor (FBS) | 35.5           |
| IDO              | Diesel engine (DE)            | 45             |
| HFO              | Fuel oil engine (FOE)         | 45             |

**Table 7**  
Electricity mix (%) in the national grid (CD4CDM, 2010; EWURA, 2007/2008; Mwakapugi et al., 2010; TANESCO, 2010; TSSED, 2010).

| Power plants          | 2000 (%) | 2015 (%) | 2020 (%) | 2026 (%) | 2030 (%) |
|-----------------------|----------|----------|----------|----------|----------|
| Hydro                 | 89       | 20       | 34       | 37       | 42       |
| Natural gas           | 9        | 53       | 48       | 33       | 28       |
| Industrial diesel oil | 2        | 4        | 3        | 2        | 2        |
| Heavy fuel oil        | 0        | 3        | 2        | 1        | 1        |
| Coal                  | 0        | 20       | 13       | 27       | 27       |

future to maintain a sustainable industrial and economic development. Also, no matter how small the electricity is generated from fossil fuel sources, it causes local as well as global pollution to the environment, thus beyond a certain point, the effects will be largely irreversible and the prospect to explore the future electricity generation would be lost forever.

## Methodology

The study mainly focuses on the top-down assessment of Tanzania's electricity production. Electricity generation from only the national grid comprising the national power plants and independent power producers (IPPs) that supply power to the national grid is considered. The study focuses on the energy requirement, net energy output, emissions to air, soil and water, abiotic resource depletion and the greenhouse gas burden of power plants. The key assumption is that power development will follow the 2009 power system master plan (PSMP) for power system expansion according to the Tanzania Electricity Supply Company Limited, TANESCO (CD4CDM, 2010; Mwakapugi et al., 2010; TANESCO, 2010).

An environmental assessment was done for Tanzania using a life cycle approach with the help of commercial software SimaPro version 7.3.3 (Pré Consultants, 2010) based on the process analysis for the years 2000, 2015, 2020, 2026 and 2030. These years have been selected according to the power system master plan which shows a significant increase in the overall power plants installed capacity (EWURA, 2007/2008; Mwakapugi et al., 2010; TANESCO, 2010). Databases used were Ecoinvent version 2.2 and the U.S. Life Cycle Inventory (USLCI) version 1.6.0 available in the SimaPro software.

**Table 8**  
Total ADP in power generation (thousand tonnes Sb-eq).

| Year                              | 2000 | 2015 | 2020 | 2026  | 2030  |
|-----------------------------------|------|------|------|-------|-------|
| Total ADP (thousand tonnes Sb-eq) | 1.4  | 56.1 | 62.8 | 105.5 | 116.2 |

Data collected were modeled to represent Tanzania's conditions, for example, using similar technologies, efficiencies and operating hours of power plants (Oeko Institute, 2010). The methodology used for life cycle impact assessment is CML 2001 (all impact categories) version 2.05. The functional unit for this study is 1 MW h net electricity at the power plant. Methodology overview in assessing environmental impacts of electricity production in Tanzania is shown in Fig. 2. It must be noted that due to limited availability of primary data, some databases as mentioned above had to be referred to. Even though efforts were made to make as reasonable a selection of data in order to simulate the situation in Tanzania, it is still a limitation which may need to be addressed in future studies.

## Results and discussion

### Power system master plan (PSMP)

The first PSMP for Tanzania was drawn in 1980 followed by others in the years 1985 and 1999. The PSMP of 2009 was initiated to guide the government and private sector in implementing a plan to meet electricity demand over the short, medium and long terms. The major challenge of the PSMP is the accelerating rate of electrification and the management of energy, for both conservation and efficiency in production, transmission and usage (Mwakapugi et al., 2010).

Some of the projects recommended in the 2009 PSMP are shown in Table 3a. Operating power plants are shown in Table 3b. The objective of the PSMP is to determine the least-cost program for the development of Tanzania's electricity generation and transmission expansion plans, and to enable TANESCO understand the existing and future level of demand for supply to customers. The 2009 PSMP and the rural electrification master plan aims to expand electrification to 20% of the population by 2015 and 50% by 2025 (Mwakapugi et al., 2010).

### Life cycle inventory (LCI)

The 2009 PSMP through the electric power sector reform act suggests that the electricity sector will undergo significant changes within the next few years. The power sector reforms aim to address the main challenges facing the Tanzanian electricity sector from electricity generation to transmission and distribution through improved electricity access by the population. From the master plan, generation capacity of the national grid is shown in Table 4, and the current government's efforts of restructuring and expanding the electricity

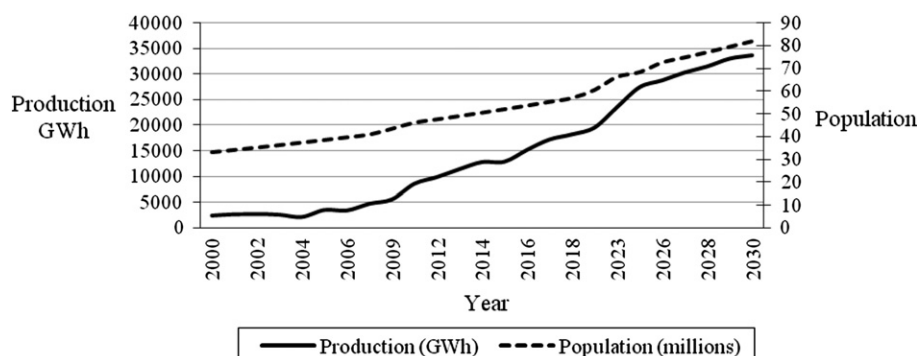


Fig. 3. Variation of projected electricity production (CD4CDM, 2010; Mwakapugi et al., 2010; TANESCO, 2010; TSSED, 2010) and population (GoT, 2010), 2000–30.



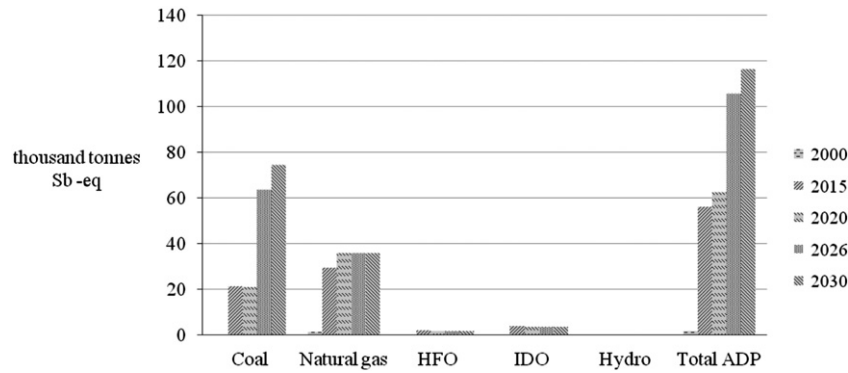


Fig. 4. Abiotic resource depletion potential (ADP) for the electricity generation, according to type of generation and total, 2000–30.

sector suggest that future electricity generation increases as shown in Table 5 for the time horizon from 2000 to 2030.

There are also plans to reduce transmission and distribution losses from the electricity grid e.g. by installing regional-wise power plants to reduce long distance transmission. Emission factors used in this study have been calculated using SimaPro version 7.3.3; Ecoinvent database version 2.2 and USLCI database version 1.6.0 (Pré Consultants, 2010). Power plant efficiency factors (Table 6) have been sourced from UNFCCC, 2011 and based on information from TANESCO.

Power plants technologies include fluidized bed combustor (coal fired power plants; coal used has relatively low sulphur content, of less than 2%), fuel oil engine (heavy fuel oil power plants), diesel engine (industrial diesel oil fired power plants), falling head run-of-river and reservoir (hydropower plants), and combined cycle, open cycle, gas engine and gas turbine (natural gas fired power plants; and only power plants with 100% natural gas firing have been considered, i.e. without co-firing of other fuels). All technologies were modelled in SimaPro version 7.3.3 to represent Tanzania (Pré Consultants, 2010). Databases were selected to represent Tanzania's conditions, for example, using similar technologies, power plant efficiencies, fuel property e.g. percent sulphur content in coal and operating hours of power plants. Electricity generation of all future thermal power plants is estimated at the average working hours of 6500 per annum and that of hydro power plants at the average of 4500 h/annum.

In the year 2000, the electricity sector depended very much on hydro which contributed 89% to the grid mix while natural gas and IDO contributed about 9% and 2% respectively. HFO and IDO power plants contributed very little and they came into operation after the severe power shedding and rationing in 2003. The 2009 PSMP suggests that there will be an increase in the share to the national grid from coal and natural gas power plants by the years 2015, 2020, 2026 and 2030 which will reduce the share of hydro and thus reduce the dependence on hydropower plants alone (Table 7).

The projected increases in electricity generation (PSMP) and population (GoT, 2010) from 2000 to 2030 are shown in Fig. 3. The share of electricity generation (for 1 MW h net electricity) from fossil fuel sources in the national grid decreases from a factor of 7.2 in 2015 to

6, 5.7 and 5.3 in 2020, 2026, and 2030, respectively using year 2000 as the base year.

#### Environmental impacts

The study shows that the overall pollutant emissions (based on the absolute annual electricity production) from power generation expansion increase significantly when comparing the years 2000, 2015, 2020, 2026 and 2030. This is because of the increased production of electricity over the years coupled with an increasing share from fossil fuels. The share of electricity generation from fossil fuels sources in the grid increases from 11% in 2000 to 80% in 2015 and decreases to 66%, 63% and 58% in 2015, 2020, 2026, and 2030 respectively (Table 7).

The 2009 PSMP suggests a significant increase in hydropower by 2030; hydropower has very low pollutant emissions compared to fossil fuel based power plants. Emissions from the hydropower mainly come from the infrastructure processes e.g. iron rod processing, gravel production, cement manufacturing (for the construction of dams), and from methane emissions from the flooded land and submerged vegetation. Although the electricity mix by 2030 will have more hydro in it in absolute terms, the overall annual burden from electricity production rises due to increased electricity generation and increased dependence on fossil fuel sources for this generation.

#### Abiotic resource depletion potential (ADP)

Abiotic resources refer to non-living natural resources, both mineral and energy. These include fossil fuels as well ores of copper, iron, etc. Abiotic resource depletion refers to the rate of deployment of the abiotic resources. The main concern of the abiotic resource depletion is the sustainability of natural resources and is expressed in kilograms of antimony equivalent, kg Sb-eq (Guinée et al., 2002).

Most of the depleted resources originate from resource consumption (fuel and other resources) for electricity generation from fossil-based fuel sources viz. coal, natural gas, IDO, HFO and very little from hydropower. The total ADP in the grid mix (Table 8 and Fig. 4) increases progressively to year 2030 when compared to year 2000 because of the significant increase in the use of fossil fuels for power generation.

The increase in overall ADP is a result of an increase in the resource consumption for the electricity generation to meet the rising electricity demand. The contribution of each power generating system to the total ADP is shown in Table 9.

Table 9

Contribution of each power generating system to the total ADP (%).

| Year        | 2000 (%) | 2015 (%) | 2020 (%) | 2026 (%) | 2030 (%) |
|-------------|----------|----------|----------|----------|----------|
| Natural gas | 71.1     | 52.4     | 57.2     | 34.1     | 30.9     |
| Coal        | 0        | 37.9     | 33.9     | 60.5     | 64.1     |
| HFO         | 0        | 3.2      | 2.8      | 1.7      | 1.5      |
| IDO         | 24.5     | 6.4      | 5.7      | 3.4      | 3.1      |
| Hydro       | 4.4      | 0.1      | 0.3      | 0.3      | 0.4      |

Table 10

Total EP from power generation, 2000–30 (thousand tonnes PO<sub>4</sub>-eq).

| Year                                           | 2000  | 2015 | 2020 | 2026 | 2030 |
|------------------------------------------------|-------|------|------|------|------|
| Total EP (thousand tonnes PO <sub>4</sub> -eq) | 0.048 | 9.0  | 9.1  | 25.9 | 30.1 |

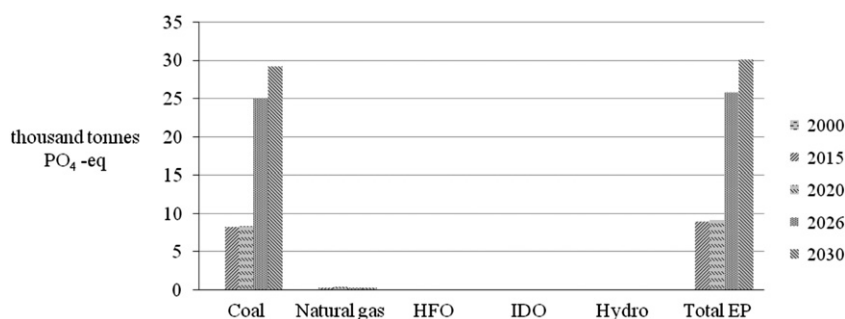


Fig. 5. Eutrophication potential (EP) for the electricity generation by power plant type and total, 2000–30.

Table 11

Relative contribution of each power generating system to the total EP (%).

| Year        | 2000 (%) | 2015 (%) | 2020 (%) | 2026 (%) | 2030 (%) |
|-------------|----------|----------|----------|----------|----------|
| Natural gas | 26.7     | 4.0      | 4.8      | 1.7      | 1.5      |
| Coal        | 0        | 92.3     | 91.2     | 96.7     | 97.1     |
| HFO         | 0        | 1.8      | 1.8      | 0.6      | 0.5      |
| IDO         | 28.8     | 1.6      | 1.6      | 0.6      | 0.5      |
| Hydro       | 44.5     | 0.3      | 0.7      | 0.4      | 0.5      |

Table 12a

Total GHG emissions from power generation, 2000–30 (million tonnes CO<sub>2</sub>-eq).

| Year                                           | 2000 | 2015 | 2020 | 2026 | 2030 |
|------------------------------------------------|------|------|------|------|------|
| Total GHG (million tonnes CO <sub>2</sub> -eq) | 0.2  | 7.2  | 8    | 13.8 | 15.2 |

#### Eutrophication potential (EP)

Eutrophication refers to the nutrient enrichment that may cause undesirable shift in biomass production and species composition in both aquatic and terrestrial ecosystems. Eutrophication can be caused by application of inorganic fertilizers in agriculture, submerged vegetation in hydroelectric reservoirs, waste water and air pollutants (Leonardo ENERGY, 2006). Thus the addition of ammonia, Chemical Oxygen Demand (COD), nitrate, nitrogen oxides and phosphate substances into the air and/or water as a result of burning of fossil fuels in power plants can contribute to eutrophication in a certain place either aquatic or terrestrial (Pré Consultants, 2010).

Submerged vegetation from flooding (hydroelectric reservoirs) may also release eutrophying substances and methane.

Most of the eutrophication originates from fossil fuel-based power plants; mostly from coal, and significant amount from natural gas, IDO, HFO and very little from hydro power plants. The EP is expressed in kilograms of phosphorus tetra oxide equivalent, kg PO<sub>4</sub>-eq (Guinée et al., 2002). The total EP in the grid mix increases progressively to year 2030 when compared to year 2000 (Table 10 and Fig. 5).

The contribution of each power generating system to the total EP is shown in Table 11. The increase in electricity generation from natural

gas and coal power plants contributes significantly to the total EP. The contribution of hydro and IDO power plants to the total EP is marginal because hydro has fewer emissions compared to other technologies while IDO and HFO has low share in the grid mix.

#### Climate change potential (CCP)

Global warming refers to the rise in global temperature caused by the increase of greenhouse gases concentrations that trap heat which would have otherwise passed out of the earth's atmosphere. Global warming has adverse impacts on ecosystems and human health, and material welfare such as increased droughts, floods, loss of polar ice caps, rise of sea-levels, change in soil moisture, loss of forests, change in wind and ocean patterns, and changes in agricultural production (Guinée et al., 2002). The main greenhouse gases are Carbon dioxide (CO<sub>2</sub>), Methane (CH<sub>4</sub>), and nitrous oxide (N<sub>2</sub>O). The GWP100 for CO<sub>2</sub>, CH<sub>4</sub>, and N<sub>2</sub>O are 1, 25, and 298 respectively (IPCC, 2007). Total greenhouse gas emissions are measured in CO<sub>2</sub> equivalents and are here referred to as climate change potential.

Most of the GHG emissions in power generation originate from fossil fuel-based power plants; in Tanzania PSMP, these are mostly from coal and natural gas, and small amount from IDO, HFO and very little from hydro power plants. The GHG emissions from power generation increases as shown in Table 12a and Fig. 6. For hydropower, methane (biogenic) contributes 51.6% of the GHG per MW h of the reservoir hydropower while infrastructure processes contribute about 48.4% (Table 12b). Recall that GHG emissions from hydropower plants come from the inclusion of infrastructure processes and methane emitted from flooded land (submerged vegetations). The percentage share of each power generating system to the total GHG emissions (Table 13) shows a significant contribution of coal and natural gas power plants due to the fact that the government is currently investing in coal and natural gas resources for the electricity production. The HFO, IDO, and hydropower plants contribute a very small percentage due to the lower share of HFO and IDO in the national grid mix as and low GHG emission intensity of hydropower.

The GHG emission per capita (Fig. 7) increases from the year 2000 to 2030 with the highest emission of 202 and 203 kg CO<sub>2</sub>-eq per

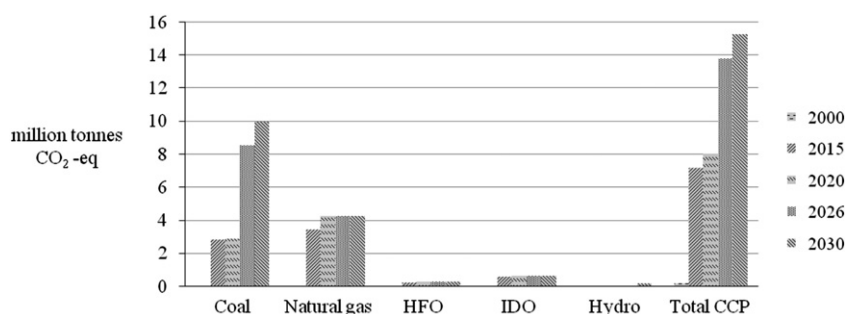


Fig. 6. Climate change potential (considering GWP with 100 years horizon) for the electricity generation by power plant type and total, 2000–30.

**Table 12b**

Contribution of methane biogenic to the total GHG from hydropower.

| Year                                                      | 2000   | 2015   | 2020   | 2026    | 2030    |
|-----------------------------------------------------------|--------|--------|--------|---------|---------|
| RES, methane biogenic, tonnes CO <sub>2</sub> -eq         | 11,456 | 12,056 | 34,533 | 57,521  | 77,954  |
| RES, infrastructure processes, tonnes CO <sub>2</sub> -eq | 10,746 | 11,308 | 32,391 | 53,953  | 73,120  |
| Total RES, tonnes CO <sub>2</sub> -eq                     | 22,202 | 23,364 | 66,924 | 111,474 | 151,074 |
| ROR, tonnes CO <sub>2</sub> -eq                           | 507    | 1980   | 2124   | 2124    | 2124    |
| Total GHG, tonnes CO <sub>2</sub> -eq                     | 22,709 | 25,344 | 69,048 | 113,598 | 153,198 |

RES—reservoir hydropower, ROR—run of river hydropower.

**Table 13**

Contribution of each power generating system to the total GHG emissions (%).

| Year        | 2000 (%) | 2015 (%) | 2020 (%) | 2026 (%) | 2030 (%) |
|-------------|----------|----------|----------|----------|----------|
| Natural gas | 59.6     | 48       | 52.7     | 30.7     | 27.8     |
| Coal        | 0        | 39.5     | 35.5     | 62       | 65.4     |
| HFO         | 0        | 3.9      | 3.5      | 2        | 1.8      |
| IDO         | 29       | 8.3      | 7.4      | 4.3      | 3.9      |
| Hydro       | 11.4     | 0.4      | 0.9      | 0.8      | 1        |

capita by the year 2024 and 2027 due to the increase of coal and natural gas use in the power production in the grid mix (34% natural gas and 28% coal in 2024 and about 31% natural gas and 30% coal in 2027).

#### Acidification potential (AP)

Acidification refers to the addition of acids on soils, groundwater, surface waters, biological organisms, ecosystems, materials and buildings. SO<sub>x</sub>, NO<sub>x</sub>, and NH<sub>3</sub> are the major acidifying pollutants. Other pollutants are hydrogen chloride and hydrogen fluoride (Pré Consultants, 2010). These pollutants react with water vapour in the atmosphere to form acids, consequently resulting in acid rain. The effect of acidification includes damages to soils, plants, animals, crumbling of building materials, and fish mortality in ponds and lakes (Guinée et al., 2002).

The acidification potential is expressed in kilograms of sulphur dioxide equivalent, kg SO<sub>2</sub>-eq (Guinée et al., 2002). Most of the AP in power generation in Tanzania originates from fossil fuel-based power plants; mostly from coal, and small amount from natural gas, IDO, HFO and very little from hydropower plants. The total AP in the grid mix increases progressively to year 2030 (Table 14 and Fig. 8).

The results also show a significant contribution of coal and natural gas power plants (Table 15) to the total AP because as per the PSMP there is an increase in the electricity generation from coal and natural gas resources. Therefore, the total AP increases due to the significant increase in the use of coal and natural gas in the national grid mix within the time horizon. The HFO, IDO, and hydropower plants contribute a

**Table 14**Total AP from power generation, 2000–30 (thousand tonnes SO<sub>2</sub>-eq).

| Year                                           | 2000 | 2015 | 2020 | 2026  | 2030  |
|------------------------------------------------|------|------|------|-------|-------|
| Total AP (thousand tonnes SO <sub>2</sub> -eq) | 0.1  | 58.4 | 66.6 | 102.7 | 111.7 |

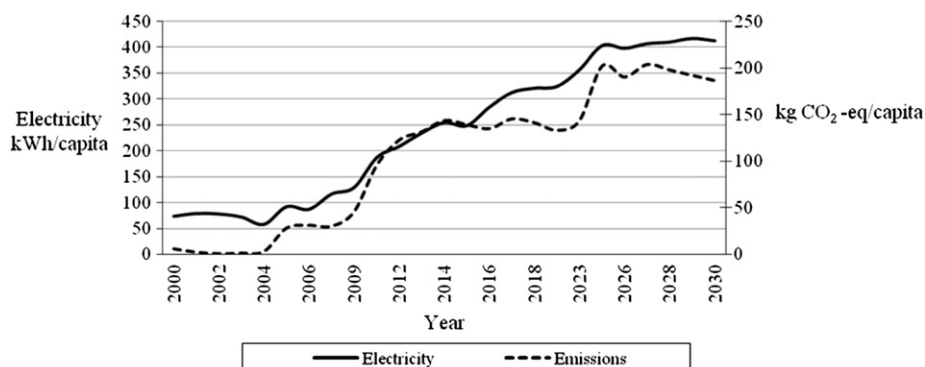
very small percentage due to the lower share of HFO and IDO in the national grid mix as well as low acidifying gas emission intensity of hydropower.

#### Environmental impact intensities

To put the results obtained in this study into perspective, the environmental impact intensities (impacts/MW h) are compared with some other countries (Table 16). The impact intensities from the literature are roughly in the same order of magnitude as for this study. For Nigeria, almost 60% of electricity is from hydro, the remaining from thermal power plants (Gujba et al., 2010). Thus, the values for all the intensities are lower than for Tanzania for the year 2015 when Tanzania is expected to have only 20% electricity from hydropower. However, the case is reverse for year 2000 when Tanzania had almost 90% electricity from hydro (Table 7). For Mexico on the other hand, more than 80% of the electricity is from fossil fuels, the remaining from hydro and nuclear (Santoyo-Castelazo et al., 2011). Thus the environmental impact intensities for Mexico are somewhat comparable for the year 2015 where the electricity mix of Tanzania has 80% fossil and 20% hydro but much higher than for year 2000 (Table 7).

Using year 2000 as a base year, fossil fuel based power plants (mainly coal, natural gas, industrial diesel oils and heavy fuel oils) would have a higher environmental impact in all the impact categories. Therefore, the significant increase of the environmental burden of electricity generation in the grid mix is a result of the increase in amount of electricity from fossil-based fuel sources.

This paper considered the environmental impact associated with PSMP projections. Alternative scenarios could involve the promotion of energy efficiency, as well as the participation of renewable power generation including hydro. Power scenarios including increased energy efficiency and renewables have been developed for many countries, e.g. by the World Wildlife Fund. One such study for Brazil was undertaken by Volpi et al. (2006). As far as we know, no similar study has been undertaken for Tanzania so far. However, considering that similar energy efficiency and renewable energy technologies are applicable in most countries, one can reasonably extend such studies to Tanzania and elsewhere. As noted earlier, there is a substantial potential for hydropower that is largely untapped. In recent years, wind power has also become cost competitive with conventional power. Energy efficiency, together with increased participation of renewables would require less generation using fossil fuels, in turn leading to lower environmental impact in all the categories considered in this study. Of course it must be noted that the social issues associated with large hydro projects, especially those involving flooding reservoirs

**Fig. 7.** Electricity production per capita (kWh/capita) and CO<sub>2</sub> emissions per capita (kg CO<sub>2</sub>-eq/capita).

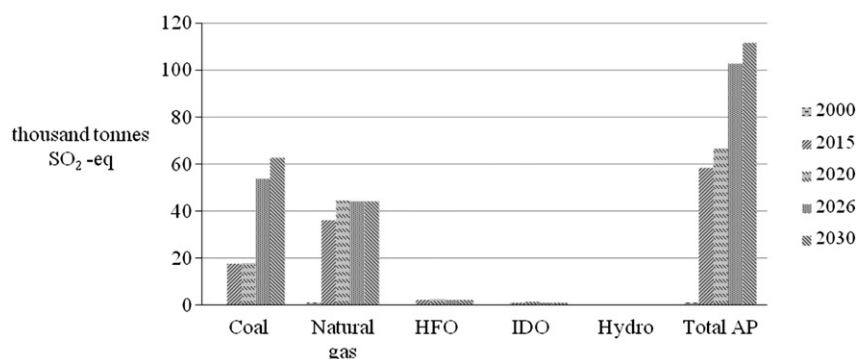


Fig. 8. Acidification potential (AP) for the electricity generation by type of power plant and total, 2000–2030.

Table 15

Contribution of each power generating system to the total AP (%).

| Year        | 2000 (%) | 2015 (%) | 2020 (%) | 2026 (%) | 2030 (%) |
|-------------|----------|----------|----------|----------|----------|
| Natural gas | 86.9     | 62.3     | 66.8     | 43.4     | 39.8     |
| Coal        | 0.0      | 30.8     | 27.0     | 52.6     | 56.3     |
| HFO         | 0.0      | 4.2      | 3.7      | 2.4      | 2.2      |
| IDO         | 10.2     | 2.6      | 2.3      | 1.5      | 1.4      |
| Hydro       | 2.9      | 0.1      | 0.2      | 0.2      | 0.3      |

Table 16

Comparison of environmental impact intensities for this study with Nigeria (Gujba et al., 2010) and Mexico (Santoyo-Castelazo et al., 2011).

| Country  | Year | CCP<br>(kgCO <sub>2</sub> -eq/<br>MW h) | AP<br>(kgSO <sub>2</sub> -eq/<br>MW h) | EP (kg PO <sub>4</sub> -eq<br>/MW h) | ADP<br>(kgSb-eq/<br>MW h) |
|----------|------|-----------------------------------------|----------------------------------------|--------------------------------------|---------------------------|
| Tanzania | 2000 | 83                                      | 0.6                                    | 0.02                                 | 0.59                      |
| Tanzania | 2015 | 560                                     | 4.53                                   | 0.7                                  | 4.36                      |
| Nigeria  | 2003 | 370                                     | 0.22                                   | 0.06                                 | 3.3                       |
| Mexico   | 2006 | 571                                     | 6.59                                   | 0.3                                  | 4.24                      |

and displacement of indigenous people have to be the considered and seriously dealt with in order to ensure that hydropower is indeed contributing to sustainable development.

## Conclusions

The overall environmental burden increases from years 2000 to 2030 because of the increase in the share of electricity generation from fossil fuels. The proposed power system master plan would have considerably higher environmental impacts compared to year 2000. The environmental burden in all categories could be reduced through increased energy efficiency and increased participation of renewables, especially hydropower, for which there is an ample potential in Tanzania, as well as other renewables such as wind power. If the power system master plan is followed about 1530 MW of hydro resource potential will still remain unused. As a result, the total generation from fossil fuels could be decreased compared to the projections presented here. Since all the environmental burdens modelled here higher for fossil fuelled power plants, the environmental impact could be reduced.

## Acknowledgments

The financial support provided by JGSEE is gratefully acknowledged. Comments by the anonymous reviewers and especially the editor-in-chief are also greatly appreciated.

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